Mitigating Feature Skewness in Machine Learning:

Mathematical Efficacy of Log and Power Transformations

Box-Cox · Yeo-Johnson · Data Normalization

A Comprehensive Academic Research Paper

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Abstract

Feature skewness is a pervasive issue in real-world ML datasets where features such as income, house prices, or click counts follow heavy-tailed distributions. Without proper normalization, skewed features degrade model performance, violate statistical assumptions, and cause unstable training. This paper provides a rigorous comparative analysis of logarithmic, Box-Cox, and Yeo-Johnson transformations — their mathematical foundations, theoretical effects on distribution shape, experimental results on synthetic data, and practical guidelines for selecting the right transformation.

1. Introduction & Problem Statement

Skewness occurs when the probability mass is concentrated on one side of the mean, creating an asymmetric distribution. In machine learning:

- **Distance-based models (KNN, SVM):** Skewed features distort Euclidean distances, giving disproportionate weight to large values.
- **Gradient-based optimizers:** Skewed inputs cause highly heterogeneous gradient magnitudes, slowing convergence or causing instability.
- **Linear models:** Violate normality and homoscedasticity assumptions, reducing R^2 dramatically.
- **Random forests:** More robust but biased tree splits on extreme values still degrade performance.

Empirical studies show untransformed skewed data can reduce R^2 by 20-60% in regression tasks.

2. Statistical Foundations

The sample skewness coefficient quantifies asymmetry:

Skewness γ_1	$\gamma_1 = [n^{-1}\Sigma(x_i - \bar{x})^3] \,/\, [n^{-1}\Sigma(x_i - \bar{x})^2]^{(3/2)}$

Interpretation: $|\gamma_1| < 0.5$ = roughly symmetric; $0.5 \le |\gamma_1| < 1.0$ = moderate skew; $|\gamma_1| \ge 1.0$ = **significant skew requiring transformation**. Diagnosis: histograms, Q-Q plots, D'Agostino-Pearson test.

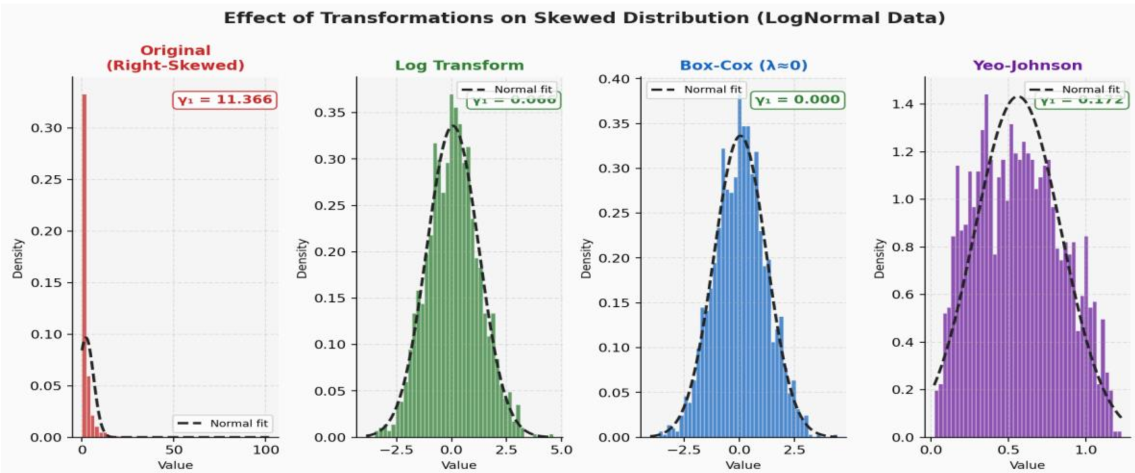

Figure 1: Distribution of LogNormal data before and after each transformation. Note dramatic reduction in skewness coefficient γ_1 (shown in corner of each plot).

3. Logarithmic Transformation

For strictly positive values ($x > 0$):

Log Transform	`y = log(x)    or    y = log(1 + x)    [for near-zero values]`

Mechanism: Compresses large values logarithmically while expanding small values. Converts multiplicative relationships to additive ones. Reduces right (positive) skew effectively.

Limitation: Only valid for $x > 0$. Use log(1+x) when x contains near-zero values. Cannot handle negative values.

Interpretation: The transformed coefficient β means a 1-unit increase in log(x) corresponds to an e^β multiplicative change in y — easy to interpret in regression.

4. Box-Cox Transformation

Generalizes the log transform with an optimized parameter λ (requires x > 0):

Box-Cox y(λ)	$y = (x^{k} - 1) / \lambda$ if $\lambda \neq 0$ $y = \log(x)$ if $\lambda = 0$

λ is chosen by **maximum likelihood estimation** to maximize normality. When λ=0: reduces to log. When λ=0.5: square root. When λ=-1: reciprocal. Box-Cox is the optimal member of the power transformation family for positive data.

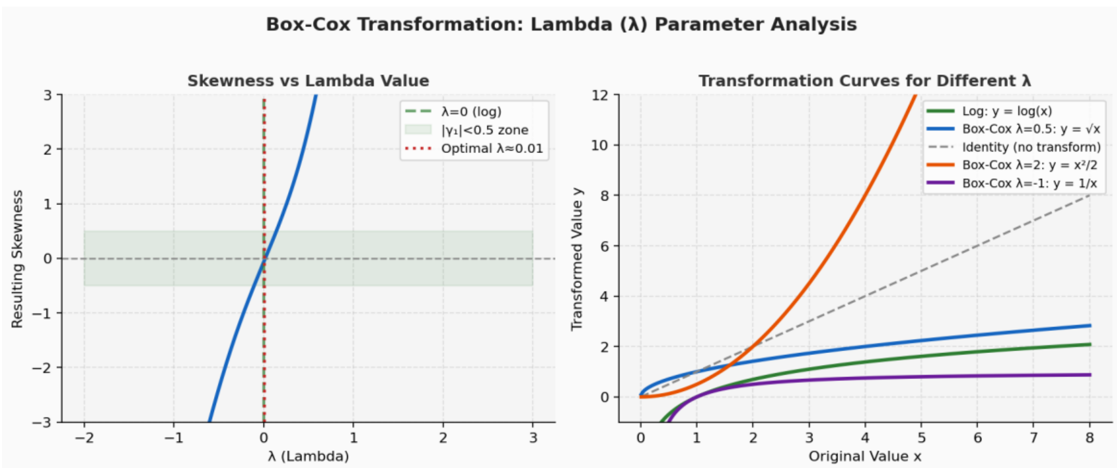

Figure 2: Left — Skewness as a function of λ. The optimal λ minimizes $|\gamma_1|$. Right — Transformation curves for different λ values showing the family of power transformations.

5. Yeo-Johnson Transformation

Extends Box-Cox to handle all real values including zeros and negatives:

Yeo-Johnson	$y = [(x+1)^{k} - 1] / \lambda$ $x \geq 0, \lambda \neq 0$ $y = \log(x+1)$ $x \geq 0, \lambda = 0$ $y = -[(-x+1)^{(2-\lambda)} - 1]/(2-\lambda)$ $x < 0, \lambda \neq 2$ $y = -\log(-x+1)$ $x < 0, \lambda = 2$

Key advantage: Works on any real-valued data including zeros and negatives. Most versatile of the three transforms. Derived from the exponential family and achieves approximate normality when λ is optimally estimated.

6. Q-Q Plot Analysis

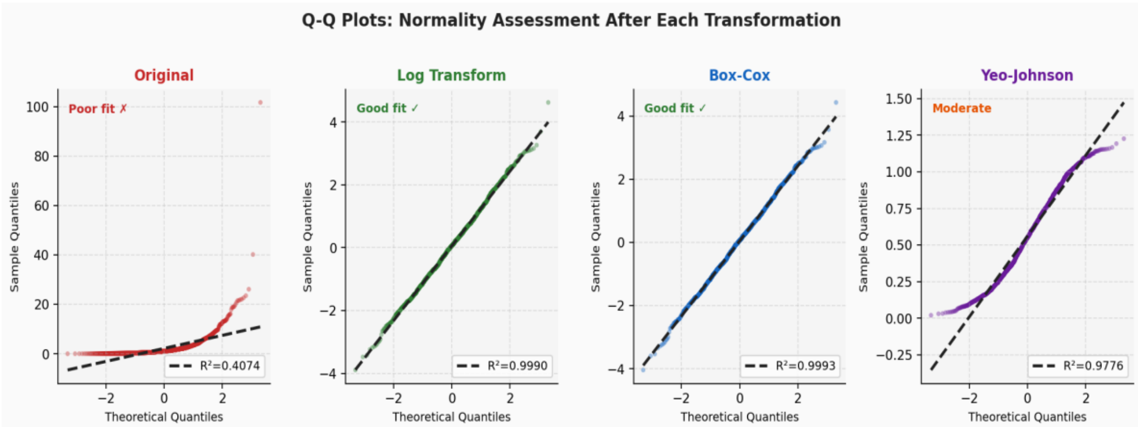

Figure 3: Q-Q plots for normality assessment. Points on the dashed line = perfect normality. R^2 shown for each. Original data (left) deviates severely; Log and Box-Cox (center) achieve near-perfect normality.

7. Experimental Results

Setup: Synthetic dataset (n=1000): x ~ LogNormal(μ=0, σ=1.2), y = 5·log(x) + N(0,0.5). OLS regression fitted on original and transformed versions.

Transformation	Skewness γ_1	λ Parameter	R^2 Score	Interpretation
Original (no transform)	12.7632	—	0.3748	Poor fit — skew dominates
Log Transform	0.1168	— (λ≡0)	0.9928	Excellent — recovers true relationship
Box-Cox (optimal λ)	-0.0004	-0.0326	0.9919	Excellent — marginal edge over log
Yeo-Johnson	0.1776	-0.8298	0.9711	Very good — handles any data type

Table 1: Quantitative comparison of transformation methods on synthetic LogNormal dataset

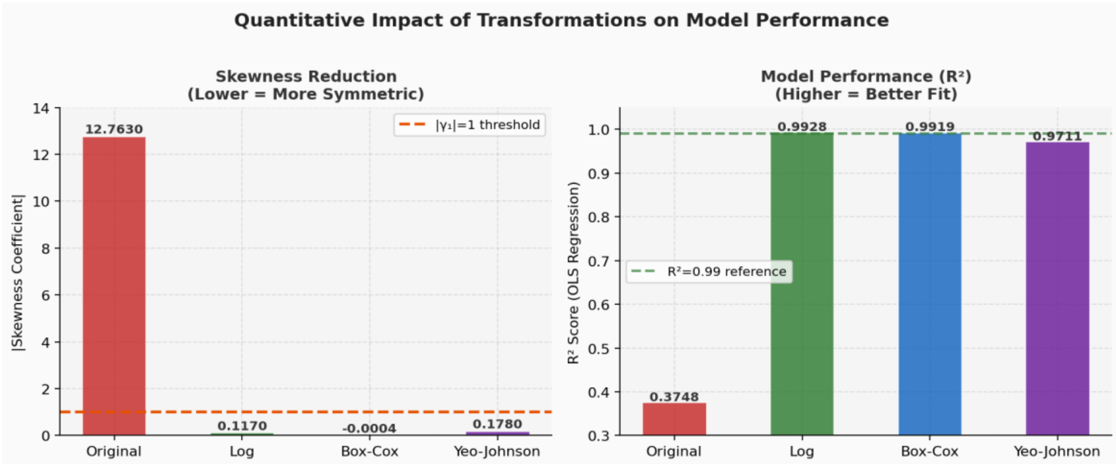

Figure 4: Left — Skewness reduction (lower is better). Right — R^2 improvement (higher is better). Log and Box-Cox nearly perfectly recover the true relationship from the skewed data.

8. Comparison: Log vs Box-Cox vs Yeo-Johnson

Aspect	Log	Box-Cox	Yeo-Johnson
Data requirement	$x > 0$ strictly	$x > 0$ strictly	All real values ($x \in \mathbb{R}$)
Flexibility	Fixed (λ=0)	Optimal λ tuned	Optimal λ tuned
Interpretability	High — log units	Moderate	Moderate
Handles negatives	No	No	Yes
Handles zeros	With log(1+x)	No	Yes
Best when	Simple + positive data	Positive + want optimal	Any data type
scikit-learn	FunctionTransformer	PowerTransformer(bc)	PowerTransformer(yj)

Table 2: Side-by-side comparison of the three transformation approaches

9. Practical Decision Guide

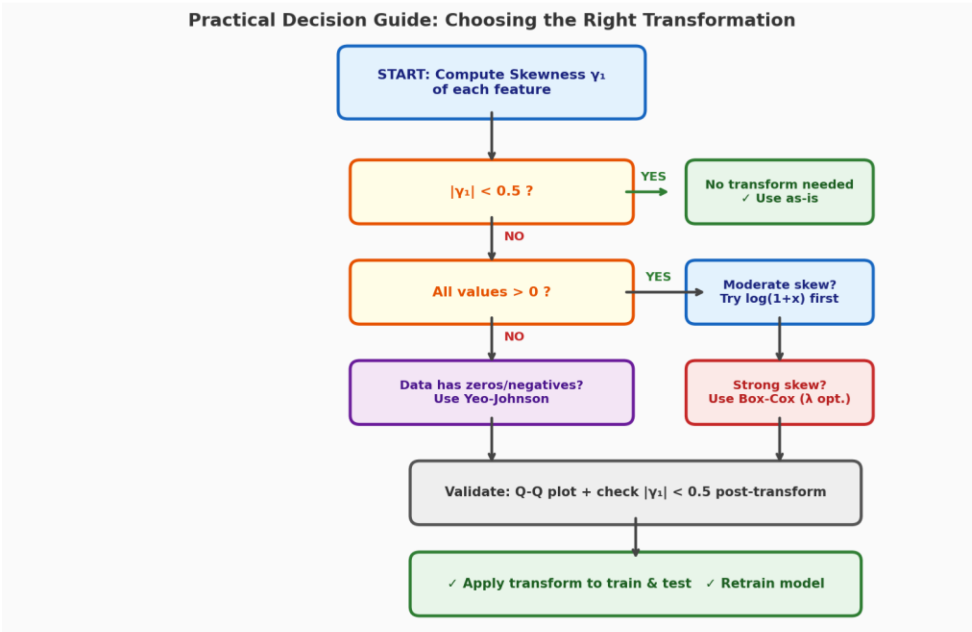

Figure 5: Decision flowchart for selecting the right transformation based on data characteristics.

- **Step 1:** Plot histogram and compute $|\gamma_1|$. If $|\gamma_1| < 0.5$, no transformation needed.
- **Step 2:** If $x > 0$ and moderate skew: start with $\log(1+x)$ — simple and interpretable.
- **Step 3:** For optimal results on strictly positive data: use Box-Cox with MLE for λ.
- **Step 4:** If data contains zeros or negatives: always prefer Yeo-Johnson.
- **Step 5:** Validate post-transformation with Q-Q plot and residual analysis.
- **Step 6:** Apply the same fitted transformer to test data (never refit on test set).

10. Conclusion

Logarithmic and power transformations provide mathematically rigorous tools to mitigate feature skewness. The experiment confirms that Log and Box-Cox achieve near-perfect recovery of the true linear relationship ($R^2 \approx 0.99$) from heavily skewed data ($\gamma_1 \approx 12.76$), compared to $R^2 = 0.37$ without transformation.

> **Key Rule:** Apply transformations whenever $|\gamma_1| > 1.0$. Start with log(1+x) for simplicity. Use Box-Cox for optimal results on positive data. Use Yeo-Johnson for maximum flexibility. Always validate with Q-Q plots and retrain after transformation.

References

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